

# **ACTIVITY SQLITE DATABASE**

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**COURSE CODE** : DSA0503

**COURSE NAME** : Query Processing for Data Science

**ACTIVITY** : SQLite Database

**1. Create the Department Table (Parent Table) with attributes Department\_ID INTEGER PRIMARY KEY, Department\_Name TEXT**

**2. Create the Student Table (Child Table) with the following attributes:**

**Student\_ID INTEGER PRIMARY KEY,  
Name TEXT,  
Age INTEGER,  
Department\_ID INTEGER,  
CGPA REAL,  
FOREIGN KEY (Department\_ID) REFERENCES Department(Department\_ID)**

**3. Display students from CSE department.**

**4. Count students in each department.**

**5. Update the foreign key. For eg., change the department of a student**

**6. Delete a department**

**7. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8.**

**CODE:**

**1. CREATING THE DATABASE**

```
import sqlite3
connection = sqlite3.connect("test.db")
cursor = connection.cursor()
print("db init")
```

**OUTPUT:** db init

**2. Create the Department Table (Parent Table) with attributes Department\_ID INTEGER PRIMARY KEY, Department\_Name TEXT**

```
cursor.execute("DROP TABLE IF EXISTS Department")
cursor.execute("""CREATE TABLE Department (Department_ID INTEGER PRIMARY
KEY, Department_Name TEXT);""")
print("Table Created Successfully")
```

**OUTPUT:** Table Created Successfully

**3. Create the Student Table (Child Table) with the following attributes:**

**Student\_ID INTEGER PRIMARY KEY,  
Name TEXT,  
Age INTEGER,  
Department\_ID INTEGER,  
CGPA REAL,  
FOREIGN KEY (Department\_ID) REFERENCES Department(Department\_ID)**

```

cursor.execute("DROP TABLE IF EXISTS Student")
cursor.execute("""CREATE TABLE Student (STUDENT_ID INTEGER PRIMARY KEY, Name
TEXT, Age INTEGER, Department_ID INTEGER, CGPA REAL, FOREIGN KEY
(Department_ID) REFERENCES Department(Department_ID));""")
print("Table Created Successfully")

```

**OUTPUT:** Table Created Successfully

#### 4. Inserting Values to the Tables

```

cursor.execute("INSERT INTO Department (Department_ID, Department_Name)
VALUES (1,'CSE'),(2,'ECE'),(3,'EEE')")
cursor.execute("INSERT INTO Student (STUDENT_ID, Name, Age,
Department_ID,CGPA) VALUES (1, 'Alice', 20, 1, 8.5), (2, 'Bob', 21, 1, 7.9),(3, 'Charlie',
22, 2, 9.1),(4, 'David', 20, 3, 8.2) ")
cursor.execute("SELECT * FROM Student;")
for student in cursor.fetchall():
    print(student)
cursor.execute("SELECT * FROM Department;")
for student in cursor.fetchall():
    print(student)

```

**OUTPUT:**

```

(1, 'Alice', 20, 1, 8.5)
(2, 'Bob', 21, 1, 7.9)
(3, 'Charlie', 22, 2, 9.1)
(4, 'David', 20, 3, 8.2)
(1, 'CSE')
(2, 'ECE')
(3, 'EEE')

```

#### 5. Display students from CSE department.

```

cursor.execute(""" SELECT Student.Name FROM Student JOIN Department ON
Student.Department_ID = Department.Department_ID WHERE
Department.Department_Name = 'CSE'; """)
print("Students in CSE Department:")
for row in cursor.fetchall():
    print(row[0])

```

**OUTPUT:**

```

Students in CSE Department:
Alice
Bob

```

#### 6. Count students in each department.

```

cursor.execute("""SELECT Department.Department_ID,
COUNT(Student.STUDENT_ID) FROM Department LEFT JOIN Student ON
Department.Department_ID = Student.Department_ID GROUP BY
Department.Department_Name;""")
for row in cursor.fetchall():
    print(f"Name: {row[0]} | Department: {row[1]} | CGPA: {row[2]}")

```

#### **OUTPUT:**

```

1: CSE: 2
2: ECE: 1
3: EEE: 1

```

### **7. Update the foreign key. For eg., change the department of a student**

```

cursor.execute("UPDATE Student SET Department_ID = 2 WHERE Name = 'Bob';")
connection.commit()
cursor.execute("SELECT * FROM Student;")
for student in cursor.fetchall():
    print(student)

```

#### **OUTPUT:**

```

(1, 'Alice', 20, 1, 8.5)
(2, 'Bob', 21, 2, 7.9)
(3, 'Charlie', 22, 2, 9.1)
(4, 'David', 20, 3, 8.2)

```

### **8. DELETE A DEPARTMENT**

```

cursor.execute("DELETE FROM Student WHERE Department_ID = 3;")
cursor.execute("DELETE FROM Department WHERE Department_ID = 3;")
cursor.execute("SELECT * FROM Department;")
for student in cursor.fetchall():
    print(student)
cursor.execute("SELECT * FROM Student;")
for student in cursor.fetchall():
    print(student)

```

**OUTPUT:**

1, 'CSE')  
(2, 'ECE')

(1, 'Alice', 20, 1, 8.5)  
(2, 'Bob', 21, 2, 7.9)  
(3, 'Charlie', 22, 2, 9.1)

**9. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8.**

```
cursor.execute("""SELECT Student.Name, Department.Department_Name,  
Student.CGPA FROM Student JOIN Department ON Student.Department_ID =  
Department.Department_ID WHERE Student.CGPA > 8.0; """)  
for row in cursor.fetchall():  
    print(f"Name: {row[0]} | Dept: {row[1]} | CGPA: {row[2]}")
```

**OUTPUT:**

Name: Alice | Dept: CSE | CGPA: 8.5  
Name: Charlie | Dept: ECE | CGPA: 9.1

[https://colab.research.google.com/drive/1gcD-M4pbdTbMM8DnDLHZPgR0JjHcw5 ?usp=sharing](https://colab.research.google.com/drive/1gcD-M4pbdTbMM8DnDLHZPgR0JjHcw5?usp=sharing)